

Advanced Scripting for Cloud Syllabus – Fall 2026

Catalog: 10-157-123
Class Section: 31689
Class Format: In Person
Location: Truax Main Building E1871



Instructor Information

Instructor: Bryan Witkowski
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Office Hours: By Appt

Course Description: Welcome to Advanced Scripting for Cloud. This course is geared to teach you how to use scripting languages and other Infrastructure as Code (IaC) tools to build and maintain cloud environments. Scripting and IaC allow you to automate the build and maintenance of cloud resources and gives you consistency, stability and economies of scale.

Course Outcomes:

- **Build/setup scripting environment for cloud providers**
- **Manage cloud services with API**
- **Utilize cloud services for monitoring and automation**
- **Explore serverless computing**
- **Deploy Infrastructure as Code (IaC) technologies to configure cloud services**

Credits: 3 **Course Format:** In Person

This is an online course where all content will be delivered asynchronously via video and slideshow instruction. Additionally, previous semester course content is available in Brightspace.

Pre-requisites: Prerequisite (grade of C or better): Python Scripting 10-152-109. Concurrent enrollment in AWS Administration 10-157-141.

Textbooks: None are required. The current pace of change in Cloud Computing makes most textbooks obsolete the moment they are published. We will utilize several online resources for reading and lab content from both Amazon and Microsoft. For a basic understanding of Amazon AWS cloud computing I would recommend the following optional book if you are interested in AWS certification:

AWS Certified Solutions Architect Official Study Guide (Associate Exam)

ISBN: 978-1-119-13855-6

Required Supplies:

1. Reliable, Fast Internet Connection and Computer
2. AWS Academy account for access to online resources and use of AWS Services.
3. GitHub account for projects and course content.

Class Workload:

Students can expect to spend 6-10 hours per week to be successful in this course. If you cannot commit to this amount of time, you may wish to re-evaluate your schedule.

Grading Policy

- All work that is to be graded must be accomplished by **your own effort**.
- Late Work: When students communicate with me ahead of missing a deadline, I am very flexible. I just need to know when you plan to finish the assigned work so I can be sure you do not fall too far behind. If you do not communicate with me about missing a deadline, you will not get credit for that assignment.

Evaluation

Assignment	Points Possible
Weekly Assignments (13)	65
Midterm Project	100
Final Project	100
Quizzes (3)	75
TOTAL POINTS	340

Grading Scale:

Grade	% Attained	Points Attained
A	92-100	312 - 340
AB	88-91	299 - 311
B	82-87	278 - 298
BC	78-81	265 - 277
C	70-77	238 - 264
D	60-69	204 - 237
F	0-59	1- 203

NOTE! Instructor reserves the right to make changes to this syllabus, course content, assignments and/or point values throughout the semester as needed. Any changes will be communicated via Brightspace and email.

Class Schedule

The following is a tentative schedule for the course. All Assignments and Labs are due at 11:59 p.m. on their due date, unless otherwise noted. Changes to this schedule can be made at any time and will be updated in Brightspace

Week	Topics	Activities, Projects and Quizzes	Due Date
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1 Aug 31 - Sept 6	Intro, Code Editors, Git/GitHub and Boto3	Setup GitHub and Boto3	Sept 6
2 Sept 7 - Sept 13	Setting up AWS Resources with Boto3	S3 Website	Sept 13
3 Sept 14 - Sept 20	Manage EC2 with Boto3	EC2 Stop/Start	Sept 20
4 Sept 21 – Sept 27	Creating Reports (pagination & filters)	Create EC2 Reports Quiz 1	Sept 27
5 Sept 28 – Oct 4	Monitoring EC2 with Boto3	Cloud Watch and SNS	Oct 4
6 Oct 5 – Oct 11	Problem Solving	Exception Handling	Oct 11
7 Oct 12 – Oct 18	Introduce Midterm Project	Midterm Project	Oct 18
8 Oct 19 – Oct 25	Getting Started with Lambda	Monitor S3 Encryption with Lambda	Oct 25
9 Oct 26 – Nov 1	Cloud Trail	Monitor Cloud Trail with Lambda	Nov 1
10 Nov 2 – Nov 8	Lambda Schedules	Stop EC2 Instances Quiz 2	Nov 8
11 Nov 9 – Nov 15	Getting Started with Terraform	Setup Terraform with AWS	Nov 15
12 Nov 16 – Nov 22	Doing More with Terraform	Deploy Lambda with Terraform	Nov 22
13 Nov 23 – Nov 29	Scripting for Azure	Setup, Install and Use	Nov 29
14 Nov 30 – Dec 6	Terraform for Azure	Setup and Use	Dec 6
15 Dec 7 – Dec 13	Introduce Final Project Monitor Security Groups with Lambda	Final Project	Dec 13
16 Dec 14 – Dec 16	Exam Week	Final Quiz	Dec 16

The instructors reserve the right to make schedule changes. All assignment due dates will be posted in the description of each assignment on Blackboard. Schedule changes may be announced in class, on blackboard, or via email. If you have any questions about due dates or the schedule, please contact the instructor during class, office hours, or via email.

Responsibilities and Policies

Syllabus Changes: As your instructor, I retain the right to make changes based on the timeline of the class, feedback from learners and/or logistical issues.

Learner Responsibilities: As a student in this class, I expect you to:

- Review and follow all Madison College policies and procedures
- Take responsibility for your own learning
- Be prepared for class and be an enthusiastic participant during class
- Treat others with tolerance and respect
- Act responsibly and reliably in group work
- Set high standards for your work
- Read the chapters and other reading assignments.
- Check Blackboard and Microsoft Teams on a regular basis.
- Complete assignments/labs/tests/quizzes/exams before due date.
- Ask questions.

Class Workload: Students can expect to spend 4-8hrs outside of the classroom to be successful in this class. If you cannot commit to this amount of time, you may wish to re-evaluate your schedule.

Instructor Responsibilities: As your instructor, I commit to communicating openly and frequently with you about this class. We will maintain a professional, safe learning environment adhering to the policies of the college. You can expect a reply to inquiries within 24-48 business hours.

Attendance: Attendance is not required for this course. It is students' responsibility to review and complete weekly content to keep up with course materials. Each week builds off previous weeks, so once a student gets behind, it is difficult to catch up.

Submitting Work: All instructions for submission of work are part of the assignments on Blackboard. Please review the assignment instructions for proper submission requirements. I recommend checking your Blackboard submissions to ensure the correct version of your documents are uploaded prior to the due date.

Student Office 365 Account: Madison College offers a student [e-mail account](#) for all students. You are responsible for monitoring your student e-mail account, Microsoft Teams and Blackboard.

Brightspace: Brightspace will be used extensively for this course for announcements, grading and delivery of class materials such as assignments & projects. Students should monitor their grades in Brightspace to assure accuracy. Your running grade total (often depicted as a letter grade) is comprised of GRADED assignments & tests, and may not reflect items which have not been graded. You can access it through the learn.madisoncollege.edu

Backup Data: You are expected to develop good computer use habits that minimize the impact of failure. I expect that you know to save your work frequently, after every major change, and to back up your work regularly. You should be able to back up all of your course files to GitHub. Remember drive failure is a matter of when ... not if.

What You Will Learn

Course Competencies (see the [course outline](#) for additional information)

1. Build/setup scripting environment for cloud providers
2. Manage cloud services with API
3. Utilize cloud services for monitoring and automation
4. Explore serverless computing
5. Deploy Infrastructure as Code (IaC) technologies to configure cloud services

School Policies and Assistance

Academic Integrity: Plagiarism, cheating and collusion are prohibited at Madison College. Plagiarism is defined as passing of another person's work as your own. Students who fail to observe these standards are subject to disciplinary action. Madison College has a strong published policy on [Academic Misconduct](#).

Equity and Inclusion: This class will be a safe and affirming learning space for all students, regardless of age, race, ethnicity, citizen status, gender, sex, sexual orientation, parental status, religion, ability, or socioeconomic status. As an instructor, I pledge to respect all students based upon these factors, including the use of preferred names and pronouns, and encourage open communication. Students are welcome and encouraged to share any/all viewpoints relevant to course material, and respectful, relevant debate is encouraged, provided all materials for the day can still be covered.

Withdrawal Policy: If a situation arises that prevents successful completion of this course, please note that it is each student's responsibility to formally withdraw from this course.

Technical Assistance: Technology resources are available for students. See the college [website](#).

Accessibility Statement: Madison College welcomes students with disabilities into the College's educational programs. Every Madison College campus has Disability Resource Services available for students with disabilities. The Disability Resource Services website provides specific information related to accommodations for disability-related barriers. For further information, please visit the <https://madisoncollege.edu/disability-resource-services>, call 608-246-6716 (Students who are deaf should use relay) or email DRStransition@madisoncollege.edu.

In order to receive consideration for reasonable accommodations for your disability, you must contact the Disability Resource Services office at the campus where you are officially enrolled. If approved for accommodations, Disability Resource Services will provide you with an accommodation plan. Please share your accommodation plan with us and discuss your approved accommodations as early in our class as possible.

Brightspace Student Support information: Visit the college support [page](#).

Brightspace Outages: Madison College does its best to keep Blackboard up and running. However, despite our best efforts, our virtual classroom may go down unexpectedly. If you cannot access our classroom, contact the [student help desk](#).

Computer difficulties are not a long-term excuse for non-participation. If you experience problems with your computer, call the computer help desk at (608) 243-4444 or toll-free at (866) 277-4445. Also, you may contact the Lab staff on the 3rd floor of the Truax Health and IT Building.

Core Workforce Skills

Local employers have consistently reported needing certain "core" skills when hiring Madison College graduates. In this course you can demonstrate these skills. Keep these in mind throughout the class—some or all may be assessed.

- SELF MANAGEMENT (manage time and stress, practice workplace etiquette, self-assess)
- CRITICAL THINKING (evaluate, solve problems, make decisions)
- SOCIAL INTERACTION (respond to feedback, respect diversity, work effectively in a team)
- COMMUNICATION (listen, speak, read, write)
- ETHICS (demonstrate acceptable behavior and judgment)